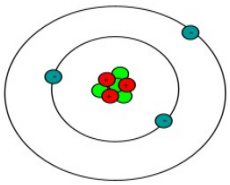
Name:_____

VALENCE ELECTRONS

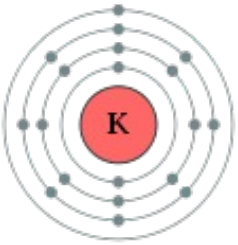
The electrons in the outermost shell of an atom are called <u>valence</u> electrons.

Here is an electron dot diagram of a lithium atom. Circle its ONE valence electron.



Look at the electron dot diagram to the right:

19: Potassium 2,8,8,1



How many electrons in the innermost shell?

How many electrons are there in the next 2 shells?

How many valence electrons does the potassium atom have?

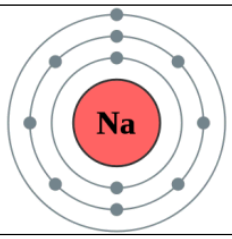
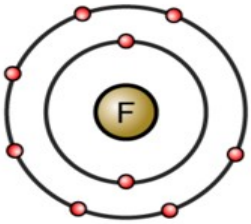
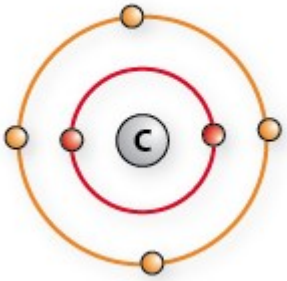
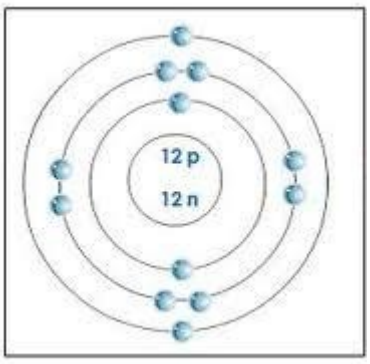
The innermost shell of every atom has a maximum of ____ electrons. Every other shell has a maximum of ____ electrons. Each shell must be full before a new one forms on top of it.

Note the electron configuration written above it. This also shows how many electrons are in each shell.

Draw the electron dot diagram and write the electron configuration for Boron with 5 electrons

How many valence electrons does boron have?

Fill in the following table

ELEMENT	DIAGRAM	HOW MANY VALENCE ELECTRONS	Electron configuration
			
			
	Carbon 		
			

Outermost shell Metals

Atoms always want to have their outermost shell **full.**
For atoms like potassium it is easy to do this by losing one electron.
Into the space below draw the electron configuration for Beryllium with 9 electrons.

How many valence electrons does beryllium have?

How can beryllium get its outermost shell full?

NOTE. Metals lose electrons to make their outermost shells full. They make positive ions.

Outermost shell Non-Metals

Into the space below draw the electron configuration for oxygen.

O_8^{16}

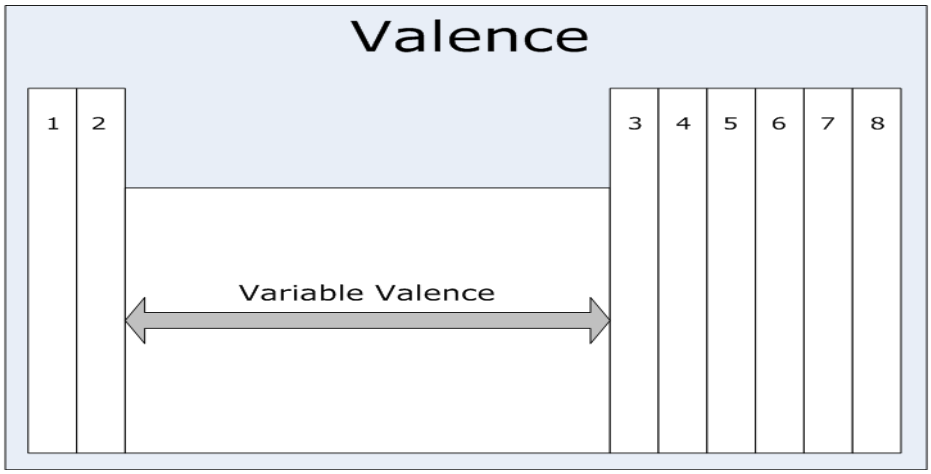
How many valence electrons does oxygen have?

How can oxygen get its outermost shell full?

NOTE. Non-Metals gain electrons to make their outermost shells full. They make negative ions.

Valence electrons on the periodic table

The periodic table is arranges so that all the elements with one valence electron are in the first column. All the elements with two electrons are in the second column. All the elements in the seventh column have _____ valence electrons.



Periodic Table of the Elements

1 H 1.008																	2 He 4.00
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.20	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.0	45 Rh 102.9	46 Pd 106.4	47 Ag 107.8	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.7	52 Te 127.6	53 I 126.9	54 Xe 131.2
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.1	74 W 183.9	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr 223.0	88 Ra 226.0	89 Ac 227.0	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (281)	111 Rg (272)	112 Uub (285)	113 Uut (284)	114 Uuq (289)	115 Uup (288)	116 Uuh (292)		

58 Ce 140.1	59 Pr 141.0	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 153.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.4	91 Pa 231.4	92 U 238.0	93 Np (237)	94 Pu (240)	95 Am (243)	96 Cm (247)	97 Bk (248)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (257)	102 No (259)	103 Lr (262)

Use what you have learned above and your periodic table to complete the following table.

element	Column of the periodic table	Number of valence electrons	Does it lose or gain electrons	Does it form positive or negative ions	Metal or non-metal
Lithium					
Magnesium					
Calcium					
Chlorine					
Carbon					
Sodium					
Bromine					
sulphur					
Iodine					

Noble Gases

Some elements have all their outer shell filled up. They are very stable and do not loose or gain electrons. They are called the noble or inert gasses.

They are found in eighth column. Find them and list them here.